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║          A Field Guide to the Six Pillars of System Change      ║
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║          From Vermont's Laboratory of Innovation                 ║
║          to National Transformation                              ║
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║          Health Transformation Review (HTR)                      ║
║          First Edition | April 2026                              ║
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*"Healthcare transformation is a systemic challenge. This book is a systemic response."*

*— Health Transformation Review, founding principle*

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**Health Transformation Review (HTR)** Vermont, USA healthtransformationreview.com

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## About This Book

This book was born from a practical observation: the people who most need to understand American healthcare transformation — hospital executives, state Medicaid directors, health economists, clinical leaders, equity researchers, operations managers, digital health investors — rarely have access to a comprehensive, integrated framework for doing so.

Policy reporters track legislation without understanding the actuarial implications. Health economists model payment reform without accounting for technology readiness. Clinical journals study care delivery innovation without attending to equity impacts. Investors screen digital health without a policy lens.

The result is a nation of specialists working with incomplete maps.

This book draws on the full analytical infrastructure of Health Transformation Review — its intelligence platform, research tools, advisory practice, academy curriculum, and seven years of deep engagement with the Vermont healthcare system — to offer something different: a complete map.

We cover all six dimensions of healthcare transformation — Policy, Economics, Technology, Clinical Innovation, Health Equity, and Operations — not as separate subjects but as an integrated system. We explain each concept clearly enough for a newcomer and rigorously enough for an expert. We use Vermont's healthcare system — simultaneously the most innovative and most financially stressed in the nation — as our primary case study.

This book is for anyone who needs to understand where American healthcare is going, why it is so hard to get there, and what the tools of transformation actually look like in practice.

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## A Note on the Vermont Case Study

Vermont occupies a unique position in American healthcare. With a population of 645,000 — smaller than most major cities — it has become the most ambitious laboratory for healthcare transformation in the United States. It was the first state to attempt single-payer healthcare (Green Mountain Care, 2011–2014). It runs the nation's most advanced all-payer global budget model (the AHEAD Model). It has some of the highest quality scores and some of the most financially stressed hospitals in the country — simultaneously.

Vermont is not a typical state. But the forces it is grappling with — aging demographics, rural hospital financial distress, commercial premium unaffordability, workforce shortages, and the structural limitations of fee-for-service payment — are national forces. Vermont is simply encountering them earlier, and more acutely, than most.

Throughout this book, Vermont appears as a case study for every major concept: not because Vermont has all the answers, but because it has been forced to ask all the right questions first.

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# TABLE OF CONTENTS

**Preface: The Incomplete Map Problem**

## PART I: UNDERSTANDING THE SYSTEM

- **Chapter 1:** The American Healthcare Crisis — A Structural Problem
- **Chapter 2:** The Six-Pillar Framework — A New Map for Complex Territory
- **Chapter 3:** Measuring Transformation — The Health Transformation Index

## PART II: THE SIX PILLARS

- **Chapter 4:** Policy — The Rules That Govern Everything
- **Chapter 5:** Economics — Following the Money Through Healthcare
- **Chapter 6:** Technology — Digital Infrastructure for the New Healthcare

- **Chapter 7:** Clinical — Where Transformation Meets the Patient
- **Chapter 8:** Health Equity — The Cross-Cutting Imperative
- **Chapter 9:** Operations — The Execution Test Every Reform Must Pass

---

## PART III: THE VERMONT LABORATORY

- **Chapter 10:** Vermont's Healthcare Crisis and the Act 167 Mandate
- **Chapter 11:** The AHEAD Model — America's Most Ambitious Global Budget
- **Chapter 12:** Vermont's Transformation Architecture — All Six Pillars in Action

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## PART IV: TOOLS FOR TRANSFORMATION

- **Chapter 13:** Interactive Analytics — The Research Lab's 19 Tools
- **Chapter 14:** Learning for Transformation — The HTR Academy
- **Chapter 15:** AI and Healthcare Intelligence — The RAG-Powered Analyst

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## PART V: THE PATH FORWARD

- **Chapter 16:** Advisory Infrastructure — From Analysis to Action
- **Chapter 17:** The Future of Healthcare Transformation — 2026 and Beyond

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**References by Chapter**

**Comprehensive Index**


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# PREFACE: THE INCOMPLETE MAP PROBLEM

Imagine you are the CFO of a regional health system in Vermont. It is January 2026. Your hospital operated at a -2.8% margin last year. Nine of Vermont's fourteen hospitals lost money in 2023. The state is negotiating a global budget that will cap your revenue growth at 3.5% annually for the next five years. A new federal Medicaid prior authorization rule just took effect, and your revenue cycle team is still figuring out what it means. You have three open primary care physician positions that have been unfilled for eighteen months. And the state legislature is debating a reference-based pricing bill that, if passed, would reduce your commercial revenue by an estimated $14 million.

You are, by any objective measure, a sophisticated healthcare professional. You have an MBA and twenty years of experience. And yet the information you need to make the next decision — any of a dozen decisions that land on your desk this week — spans at least five distinct domains:

- **Policy** (what does the legislation actually say, and what does CMS allow?)
- **Economics** (what are the financial implications of a 3.5% growth cap at your current cost structure?)
- **Technology** (can your EHR system produce the data required by the new global budget reporting requirements?)
- **Clinical** (what care delivery changes would actually bend the cost curve while maintaining quality?)
- **Equity** (does your global budget design create perverse incentives to avoid high-need, high-cost patients who are disproportionately rural and low-income?)

No single expert, publication, or database gives you all of this in one place. You are working with an incomplete map.

This book is about building the complete map.

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Healthcare transformation is not a policy problem, an economics problem, a technology problem, a clinical problem, or an equity problem. It is all of them simultaneously, and the interactions between them are where transformation succeeds or fails.

The AHEAD Model — Vermont's global budget — is a policy innovation. But it fails as a cost-containment mechanism unless the economics work (benchmark methodology must reflect actual cost drivers), the technology exists (all-payer claims data infrastructure must be robust), the clinical delivery system changes (care management programs must reduce preventable utilization), and the equity design is sound (budget allocations must not incentivize patient avoidance among high-need populations).

The individual who understands only one dimension of this system will make decisions that seem correct from within their specialty and produce unintended consequences in every other dimension.

This is the problem Health Transformation Review was founded to solve — and this book is the most complete expression of that work.

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*A note on how to read this book:*

Each chapter is designed to stand alone. You can read it cover to cover, or you can use it as a reference — going to Chapter 4 (Policy) when you need to understand Medicare Advantage regulation, or to Chapter 8 (Health Equity) when you need to understand SDOH measurement, or to Chapter 11 (AHEAD Model) when you need to understand global budget mechanics.

The chapters in Part II (The Six Pillars) are deliberately comprehensive — each one covers concepts ranging from introductory to advanced. If you are an expert in one pillar, skim the foundational sections and focus on the analytical frameworks. If you are new to a pillar, read every section; the concepts build on each other.

The Vermont chapters in Part III assume familiarity with the foundational concepts from Part II. Read those first.

The tools chapters in Part IV (Research Lab, Academy, AI) describe the HTR platform's analytical instruments. These chapters are useful whether you use the platform or not — the tools embody a set of analytical frameworks that are valuable to understand regardless of the medium.

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# PART I: UNDERSTANDING THE SYSTEM

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# CHAPTER 1: The American Healthcare Crisis — A Structural Problem

## The Numbers That Demand Explanation

Every serious analysis of American healthcare begins with the same set of facts, because they are so extraordinary that they demand explanation before anything else can make sense:

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┌──────────────────────────────────────────────────────────────┐
│       U.S. HEALTHCARE SPENDING vs. OUTCOMES (2024)           │
├───────────────────────────────┬──────────────────────────────┤
│  Per-capita spending          │  $14,570 (est. 2024)         │
│  OECD peer average            │  ~$6,200                     │
│  U.S. premium above peers     │  ~135%                       │
├───────────────────────────────┼──────────────────────────────┤
│  Life expectancy (U.S.)       │  76.4 years (2023)           │
│  Life expectancy (Japan)      │  84.3 years                  │
│  Life expectancy (Australia)  │  83.2 years                  │
│  Life expectancy (Germany)    │  80.6 years                  │
├───────────────────────────────┼──────────────────────────────┤
│  Preventable mortality rank   │  Last among 11 peer nations  │
│  Infant mortality rank        │  Last among 11 peer nations  │
│  Access rank                  │  Last among 11 peer nations  │
│  Equity rank                  │  Last among 11 peer nations  │
└───────────────────────────────┴──────────────────────────────┘
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Source: Commonwealth Fund International Health Policy Survey, 2024;
CMS National Health Expenditure Accounts, 2024
```

These numbers are not the result of insufficient spending. The United States spends roughly twice what peer nations spend per person on healthcare, and achieves worse outcomes on nearly every measure that matters — life expectancy, preventable death, infant mortality, and healthcare equity.

This is not a resource problem. **It is a structural problem.**

Understanding what that means — and what it does not mean — is the starting point for everything in this book.

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## What "Structural" Actually Means

When analysts say American healthcare has a structural problem, they mean that the rules, incentives, and organizational arrangements of the system produce these poor-value outcomes as a predictable consequence of how the system is designed — not as a failure to follow the design.

Consider the following structural features:

**1. Fee-for-service payment**

The dominant payment model in American healthcare pays providers for the volume of services delivered — each test ordered, each procedure performed, each visit completed. A physician who orders an unnecessary MRI gets paid for it. A hospital whose readmission rate is high gets paid for each readmission. A health system that manages chronic disease poorly, generating frequent hospitalizations, is paid for every hospitalization.

There is no mechanism in fee-for-service payment to reward preventing illness, coordinating care across providers, reducing unnecessary utilization, or improving population health. The incentive structure is perfectly aligned with high cost and poorly aligned with good outcomes.

**2. Fragmented delivery**

American healthcare is delivered by thousands of independent organizations — hospitals, physician groups, post-acute facilities, behavioral health providers, pharmacy benefit managers — that typically do not share data, coordinate care, or operate under a unified accountability structure for any patient's overall health.

When a patient with Type 2 diabetes, hypertension, and depression sees a primary care physician, an endocrinologist, a cardiologist, and a therapist — none of whom share records or communicate with each other — the care is fragmented by design. There is no one accountable for the whole patient.

**3. Inadequate attention to social determinants**

A person's health is determined by many more factors than the healthcare they receive. Housing stability, food security, income, education, social connection, environmental quality, and access to transportation — the social determinants of health (SDOH) — account for an estimated 30-55% of health outcomes. Yet the U.S. healthcare system is organized almost entirely around clinical interventions that address only the downstream consequences of social conditions it does not attempt to change.

A patient who is hospitalized for a diabetic crisis, discharged, and returns to food insecurity and an unheated apartment will likely be readmitted within 30 days. No amount of clinical excellence at the time of discharge changes that.

**4. Inequitable access by design**

American healthcare access is determined by insurance status, which is determined by employment status, income, and geography — structural factors that are not distributed equally across the population. Uninsured rates are higher among Black, Hispanic, and Native American populations, rural populations, and low-income populations. These are the same populations with the highest rates of chronic disease and the greatest need for healthcare — and the least consistent access to it.

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## The Cost of the Structure

The structural features described above generate the spending-outcomes gap in a predictable way:

```
┌──────────────────────────────────────────────────────────────┐
│       HOW STRUCTURAL PROBLEMS PRODUCE THE OUTCOMES GAP       │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  FEE-FOR-SERVICE PAYMENT                                     │
│     → Incentivizes volume over value                         │
│     → Rewards procedures, not prevention                     │
│     → No accountability for total cost of care               │
│         ↓                                                    │
│  FRAGMENTED DELIVERY                                         │
│     → Duplicated testing, contradictory prescribing          │
│     → Preventable readmissions (~$26B annually)              │
│     → High administrative overhead (~$500B annually)         │
│         ↓                                                    │
│  INADEQUATE SDOH INVESTMENT                                  │
│     → Preventable chronic disease progression                │
│     → High-cost emergency utilization for social needs       │
│     → Poor adherence due to unaddressed social barriers      │
│         ↓                                                    │
│  INEQUITABLE ACCESS                                          │
│     → Delayed care → higher acuity, higher cost              │
│     → Emergency department as primary care                   │
│     → Worse outcomes, especially in rural/minority pop.      │
│         ↓                                                    │
│  RESULT: $14,570/capita with last-place outcomes             │
│                                                              │
└──────────────────────────────────────────────────────────────┘
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The key insight is that these are not individual failures — they are system properties. A single hospital cannot fix fee-for-service payment. A single physician cannot eliminate fragmentation. A single health plan cannot address food insecurity. **Structural problems require structural solutions.**

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## The Transformation Imperative

Healthcare transformation is the collective term for the set of structural reforms that aim to address these foundational problems. It is not one initiative or one policy — it is a multi-decade, multi-dimensional project of redesigning the rules, incentives, and organizational arrangements of the American healthcare system.

The transformation agenda includes:

- **Payment reform:** Shifting from fee-for-service to alternative payment models (APMs) that reward value, outcomes, and total cost of care reduction
- **Care delivery redesign:** Organizing care around patients rather than providers, enabling team-based, coordinated, population health-oriented delivery
- **Technology investment:** Building the digital infrastructure — interoperable EHRs, real-time data exchange, AI-powered analytics — that makes population health management possible
- **Equity-centered design:** Ensuring that every reform initiative explicitly addresses disparities and invests in the social determinants of health
- **Administrative simplification:** Reducing the $500 billion annual administrative burden that consumes resources without improving outcomes

None of these reforms operates in isolation. Payment reform does not work without technology readiness. Technology investment does not improve outcomes without care delivery redesign. Care delivery redesign fails the populations that need it most if equity is not designed in from the start. And all of it fails if the operations infrastructure cannot execute.

This is why a six-pillar framework — and not a simpler, more linear approach — is necessary to understand and advance healthcare transformation.

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### The Scale of Change Required

It is worth pausing to appreciate the scale of the challenge. American healthcare is:

- A $4.5 trillion industry (2023)
- Employing approximately 22 million people
- Organized across approximately 6,000 hospitals, 1 million licensed physicians, 5,000+ health plans, and hundreds of thousands of ancillary providers
- Governed by a complex web of federal and state laws, regulations, and payment rules
- Culturally embedded in a society with deeply held beliefs about the doctor-patient relationship, medical autonomy, and the limits of government involvement in healthcare

Transforming this system is not a project. It is a generation-long civilizational effort. Understanding it — really understanding it, in its full complexity — requires the kind of multi-dimensional framework this book provides.

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### Key Concepts Introduced in This Chapter

> **Fee-for-service (FFS):** A payment model in which providers are paid for each service delivered, regardless of whether the service improves the patient's health or represents appropriate use.

> **Alternative payment model (APM):** A payment arrangement that ties reimbursement to quality, outcomes, or total cost of care rather than volume. Includes bundled payments, shared savings, global budgets, and capitation.

> **Social determinants of health (SDOH):** The non-clinical factors that shape health outcomes: housing, food security, income, education, transportation, social connection, and environmental quality.

> **Healthcare transformation:** The multi-dimensional, long-term project of restructuring American healthcare's payment models, care delivery systems, technology infrastructure, equity practices, and administrative operations to produce better outcomes at lower cost.

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# CHAPTER 2: The Six-Pillar Framework — A New Map for Complex Territory

### Why Single-Lens Analysis Fails

In 2019, a major regional health system in the Mid-Atlantic decided to enter its first Medicare Shared Savings Program (MSSP) contract. The decision was made primarily by the economics team, which modeled projected shared savings based on the organization's historical medical cost trend. The model was rigorous and the projections were reasonable.

What the economics team did not know — because no one asked the technology team — was that the health system's EHR did not have the attribution functionality required to identify which patients were in the MSSP population. They could not run the care gap analyses that drive shared savings performance. They did not discover this until six months after the contract start, when they missed the first care quality reporting deadline.

The economics were right. The technology was wrong. The result was a financial loss.

This is not a story about incompetent people. It is a story about what happens when a complex, multi-dimensional decision is analyzed through a single lens.

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### The Six-Pillar Framework: An Introduction

Health Transformation Review's Six-Pillar Framework holds six analytical dimensions in view simultaneously. Each pillar represents a distinct domain of expertise, a set of analytical questions, and a body of evidence. Together, they constitute a complete map of healthcare transformation.

**THE SIX-PILLAR FRAMEWORK**

| | |
|---|---|
| POLICY (20% weight) | The legislative, regulatory, and payment reform environment that sets the rules for all transformation |
| ECONOMICS (25% weight) | The financial flows, market structures, and incentives that drive behavior across the system |
| TECHNOLOGY (15% weight) | The digital infrastructure, clinical information systems, AI tools, and interoperability standards |
| CLINICAL (20% weight) | Care delivery, quality measurement, workforce dynamics, and the design of care models |
| EQUITY (10% weight) | Systematic differences in health outcomes and access by race, income, geography, and other characteristics |
| OPERATIONS (10% weight) | The administrative, operational, and execution infrastructure that determines whether transformation can be executed |

The weights shown above reflect their relative importance in the Health Transformation Index (described in Chapter 3). They are not a ranking of importance — all six pillars are essential. The weights reflect the relative measurability and pace of change of each dimension.

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## Pillar 1: Policy

**The question Policy answers:** What are the rules of the game, and how are they changing?

Policy is the foundational pillar because payment policy is the primary lever for system-level transformation. You cannot change how care is delivered without changing how care is paid for. You cannot change how care is paid for without changing the policy environment that determines payment rules.

The policy environment for healthcare transformation operates at three levels:

**Federal level:**

- The Affordable Care Act (ACA) of 2010 — still the most transformative piece of healthcare legislation in a generation, establishing the legal and financial architecture for coverage expansion and insurance market reform
- CMS payment rules — the annual IPPS (Inpatient Prospective Payment System), OPPS (Outpatient), and QPP (Quality Payment Program) rules that govern how hospitals and physicians are paid under Medicare
- CMMI (Center for Medicare and Medicaid Innovation) — the federal laboratory for payment model innovation, testing APMs, bundled payments, global budgets, and accountable care models
- No Surprises Act, Consolidated Appropriations Act, Inflation Reduction Act — legislation continuing to reshape payment and pricing

**State level:**

- Medicaid waiver programs — Section 1115 waivers allow states to experiment with Medicaid eligibility, benefits, and payment beyond standard federal rules; Section 1115A waivers (used by Vermont's AHEAD Model) allow global budget experiments with Medicare participation
- State innovation programs (SIM grants, all-payer model legislation)
- Price transparency requirements, prior authorization reform, and scope of practice legislation

**Market level:**

- Private payer contracts — the contracts between health plans and providers that determine actual payment rates for the majority of Americans
- Employer benefit design — large self-insured employers that directly determine healthcare benefits for millions of workers

Understanding the policy environment requires tracking all three levels simultaneously. A federal payment rule change may have opposite effects in states with strong innovation programs versus states with minimal regulatory infrastructure.

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## Pillar 2: Economics

**The question Economics answers:** Who pays, how much, and for what — and how does that change behavior?

Economics is the most heavily weighted pillar in the HTR framework because financial incentives are the most powerful behavioral lever in healthcare. Organizations respond to how they are paid. Physicians prescribe more when paid per service. Hospitals invest in profitable service lines and disinvest from unprofitable ones. Payers design benefits to attract healthy enrollees and deter sick ones.

The economics of healthcare transformation center on the shift from fee-for-service payment to **value-based care (VBC)** — arrangements in which payment is tied to quality, outcomes, and total cost of care rather than volume.

```
FEE-FOR-SERVICE ECONOMICS
─────────────────────────
Revenue = Volume × Unit Price
Incentive: More services = More revenue
Risk: None (payer absorbs all utilization risk)
Cost control: None built in

VALUE-BASED CARE ECONOMICS
───────────────────────────
Revenue = Capitation/Global Budget − Cost of Care + Shared Savings
Incentive: Better outcomes + Lower cost = More margin
Risk: Shared between payer and provider
Cost control: Built into contract structure
```

The economics of healthcare extend far beyond the payment model, however. Health economics encompasses:

- **Cost-effectiveness analysis (CEA):** Quantifying the value of clinical interventions — what does it cost to gain one additional quality-adjusted life year (QALY)?
- **Actuarial science:** Predicting the cost of insuring a population, pricing premiums, designing benefits, and managing risk
- **Health system financial analysis:** Understanding operating margins, payer mix, capital structure, and financial sustainability for hospitals and health systems
- **Market economics:** How hospital mergers, insurance market concentration, and private equity shape prices and access

All of these dimensions are relevant to healthcare transformation because the economics determine what is financially viable. The most clinically excellent model of care cannot spread if it is not economically sustainable for the providers who deliver it.

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## Pillar 3: Technology

**The question Technology answers:** Does the digital infrastructure exist to make transformation work in practice?

Technology is simultaneously the greatest enabler of healthcare transformation and a source of its most persistent barriers. The shift to value-based care requires capabilities that are simply impossible without sophisticated digital infrastructure:

- **Population identification:** To manage the health of a defined population, you must be able to identify all the patients in that population, track their clinical status, and flag care gaps in real time. This requires interoperable data exchange across all the providers who touch those patients.
- **Attribution:** Under most VBC contracts, you must know which patients are "attributed" to your organization for cost and quality accountability purposes. Attribution algorithms depend on claims data and clinical data that must flow consistently and accurately.
- **Quality measurement:** HEDIS, CMS Stars, MIPS, and every other quality measurement program requires structured clinical data (lab values, diagnoses, procedures, medications) extracted systematically from EHR systems and submitted to measurement entities.
- **Risk stratification:** Identifying which patients are at risk of becoming high-cost, high-utilization is a predictive analytics problem that requires machine learning on large clinical and claims datasets.
- **Care coordination:** Ensuring that a patient with five chronic conditions and five different providers has coherent care requires real-time data sharing across those providers — which requires interoperability standards.

The central technology standard for healthcare interoperability is **FHIR** (Fast Healthcare Interoperability Resources), the HL7 standard for exchanging healthcare information electronically. FHIR defines a set of data resources (Patient, Condition, Observation, MedicationRequest, etc.) and a RESTful API specification for exchanging them. Since 2020, CMS has required all hospitals and health plans to implement FHIR-based Patient Access and Provider Access APIs under the 21st Century Cures Act.

**Artificial intelligence** is reshaping healthcare at every level — from diagnostic imaging analysis to sepsis prediction to administrative automation. But AI in healthcare introduces a specific category of risk that does not exist in other industries: **algorithmic bias**. A predictive model trained on historical clinical data will encode the disparities in that data. A sepsis prediction algorithm trained on a predominantly white patient population may perform worse for Black patients. An insurance underwriting model trained on biased historical claims may discriminate against marginalized groups. Technology governance — including bias testing, explainability requirements, and ongoing monitoring — is not optional; it is foundational.

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## Pillar 4: Clinical

**The question Clinical answers:** How does care actually change at the point of patient contact?

All the policy reform, economic restructuring, and technology investment in the world ultimately aims to change one thing: how healthcare is delivered to patients, and whether it improves their health. The Clinical pillar is where the other five pillars are either validated or exposed as insufficient.

Clinical transformation encompasses several interconnected domains:

**Care delivery model redesign:** The fundamental unit of American primary care — the 15-minute physician visit organized around acute complaints — is not designed for chronic disease management, prevention, or population health. Effective clinical transformation requires new models:

- **Patient-Centered Medical Home (PCMH):** A team-based primary care model emphasizing care coordination, preventive care, and chronic disease management
- **Accountable Care Organization (ACO):** A network of providers taking collective accountability for the cost and quality of care for a defined patient population
- **Integrated Behavioral Health:** Embedding mental health and substance use disorder treatment in primary care settings
- **Hospital at Home:** Delivering acute care in patients' homes for eligible conditions

**Clinical quality measurement:** Quality in healthcare is measured systematically through programs including:

- **HEDIS** (Healthcare Effectiveness Data and Information Set): 90+ measures spanning preventive care, chronic disease management, and behavioral health, maintained by NCQA
- **CMS Star Ratings:** 1–5 star ratings for Medicare Advantage and Part D plans based on clinical quality, patient experience, and administrative performance
- **MIPS** (Merit-based Incentive Payment System): The quality-payment program for Medicare physicians, combining quality measures, improvement activities, and interoperability performance
- **Joint Commission accreditation:** Hospital quality and safety standards that influence both regulatory status and payer contracting

**Workforce:** Healthcare transformation's most underappreciated constraint is workforce. The physicians, nurses, advanced practice providers, pharmacists, community health workers, and care coordinators required to deliver transformed care models are not a fixed resource — they must be trained, recruited, retained, and organized in new ways.

The United States faces significant workforce shortfalls:

- Primary care physician shortage: HRSA projects a deficit of 68,000 primary care physicians by 2035
- Nursing: RN vacancy rates reached 17% nationally in 2023
- Behavioral health: >70% of counties in the U.S. lack adequate mental health providers

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## Pillar 5: Equity

**The question Equity answers:** Who is left out, and how do we design systems that serve everyone?

Health equity is the most complex pillar conceptually, and the most important cross-cutting lens analytically. It is not a separate subject from the other five pillars — it is a dimension that must be applied to each of them.

A policy reform that is racially neutral in design may produce racially inequitable outcomes if it requires administrative steps (enrollment processes, documentation requirements) that are more burdensome for low-income, minority, or non-English-speaking populations. An economic reform that generates system-wide cost savings may widen health disparities if the savings come from reducing services used disproportionately by high-need populations. A technology tool that is 95% accurate on average may be 80% accurate for the minority population that is the primary target of a clinical intervention.

**Health disparities** exist along multiple dimensions:

- **Race and ethnicity:** Black Americans have a life expectancy 5 years shorter than white Americans; have higher rates of hypertension, diabetes, maternal mortality, and infant mortality; and face documented disparate treatment in clinical settings
- **Income:** People in the lowest income quintile have a life expectancy 15 years shorter than people in the highest income quintile
- **Geography:** Rural Americans have higher rates of premature death from all causes, lower access to primary and specialty care, and higher rates of poverty and social isolation
- **Sexual orientation and gender identity:** LGBTQ+ individuals face higher rates of mental health conditions, substance use, and chronic disease, often driven by social stress and inadequate access to affirming care

**Social determinants of health (SDOH)** are the structural conditions that produce these disparities. Housing instability, food insecurity, poverty, lack of education, environmental pollution, community violence — these are the upstream causes of the downstream health disparities that the clinical system sees as disease. Healthcare transformation that ignores SDOH is treating symptoms rather than causes.

**Equity-weighted analysis** — the practice of giving additional analytical value to interventions that disproportionately benefit disadvantaged populations — is an emerging methodology that bridges the equity and economics pillars. The HEROI (Health Equity Return on Investment) metric, used in HTR's Health Equity Studio, adjusts cost-effectiveness calculations to reflect the additional social value of reducing health disparities.

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## Pillar 6: Operations

**The question Operations answers:** Can this transformation actually be executed at scale?

Operations is the pillar that transforms plans into reality — or exposes them as fantasies. A transformation strategy that is legally permissible (Policy), economically funded (Economics), technologically capable (Technology), clinically proven (Clinical), and equitably designed (Equity) can still fail completely if the operational infrastructure cannot execute it.

The operations pillar encompasses:

**Revenue cycle management (RCM):** The end-to-end process of converting clinical services into revenue — patient registration, insurance verification, charge capture, claim submission, denial management, and collections. Revenue cycle is the financial nervous system of a health system; under value-based care, it must evolve from a volume-counting machine to a value-demonstrating engine.

Key revenue cycle metrics:

- **Days in accounts receivable (AR):** How long it takes to collect after a service is delivered. Industry median: 45–55 days. Under global budget models, this metric changes character entirely.
- **Denial rate:** Percentage of submitted claims initially denied by payers. Industry benchmark: <5%. Average for hospitals entering VBC: often 8–12%.
- **Clean claim rate:** Percentage of claims submitted without errors. Benchmark: >95%.
- **Net collection rate:** Dollars collected as a percentage of net allowable charges. Benchmark: >95%.

**Workforce operations:** The administrative and operational aspects of workforce management — credentialing, onboarding, scheduling, turnover management, labor relations, and staffing model design. Workforce operations is distinct from clinical workforce (the clinical competency of the workforce) and workforce policy (scope-of-practice legislation). It is the machinery by which the right clinician gets the right credential to work at the right facility at the right time.

Credentialing cycle time — the time from a new provider's hire date to their first billable encounter — is a critical operations metric that is rarely analyzed in healthcare transformation literature. In a health system entering a new VBC contract that requires a 25% expansion of care management staff, a credentialing cycle time of 90 days creates a 90-day revenue gap and a 90-day care gap simultaneously.

**Quality and compliance infrastructure:** The accreditation, audit, and compliance systems that ensure a health system meets regulatory and contractual standards. Joint Commission surveys, NCQA Health Plan accreditation, HIPAA compliance programs, CMS Conditions of Participation — these are not bureaucratic overhead. They are the minimum operational standards below which transformation cannot safely proceed.

**Supply chain and facilities:** In the context of healthcare transformation, supply chain is most relevant when care delivery models change. When hospitals shift from inpatient to outpatient care models, or from clinic-based to home-based care, the procurement patterns, inventory management requirements, and facility utilization profiles change fundamentally. A supply chain designed for a high-inpatient-volume model may be the wrong supply chain for an ACO focused on community-based care.

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## How the Pillars Interact: The Vermont AHEAD Model

Vermont's AHEAD (All-payer Health Equity Approaches and Development) model — the most ambitious global budget experiment in the United States — illustrates how the six pillars interact in practice:

```
┌──────────────────────────────────────────────────────────┐
│              VERMONT AHEAD MODEL: SIX-PILLAR LENS         │
├─────────────┬────────────────────────────────────────────┤
│ POLICY      │ 1115A federal waiver structure; CMS CMMI    │
│             │ agreement; Medicare participation terms;    │
│             │ hospital global budget statute              │
├─────────────┼────────────────────────────────────────────┤
│ ECONOMICS   │ All-payer benchmark methodology; global     │
│             │ budget math; what happens when hospitals    │
```

```
│              │ exceed budget; risk corridor design           │
├──────────────┼───────────────────────────────────────────────┤
│ TECHNOLOGY   │ All-payer claims data infrastructure;         │
│              │ FHIR interoperability for care coordination;  │
│              │ population health analytics capability        │
├──────────────┼───────────────────────────────────────────────┤
│ CLINICAL     │ Care delivery changes required to bend cost   │
│              │ curve; behavioral health integration;         │
│              │ primary care capacity and access              │
├──────────────┼───────────────────────────────────────────────┤
│ EQUITY       │ Does budget mechanism incentivize avoiding    │
│              │ high-need patients? Rural equity adjustors;   │
│              │ LGBTQ+ network adequacy                       │
├──────────────┼───────────────────────────────────────────────┤
│ OPERATIONS   │ Revenue cycle capacity for attribution;       │
│              │ credentialing pipeline for care coordination  │
│              │ workforce; supply chain for care migration    │
└──────────────┴───────────────────────────────────────────────┘
```

A policy analyst who examines only the waiver language will miss the actuarial problems. An economist who models only the financial flows will miss the clinical implementation barriers. An operations analyst catches that the revenue cycle cannot handle the attribution methodology before the contract is signed.

This is why the six-pillar framework exists — not as an academic taxonomy, but as a practical discipline for avoiding costly analytical blind spots.

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## The Framework in Practice: Three Decision Types

The six-pillar framework is not merely descriptive. It is a decision-support structure. Here is how it applies to three of the most common decision types in healthcare:

### Decision Type 1: Should We Enter a Risk Contract?

A regional health system is considering its first full-risk ACO REACH contract. The six-pillar checklist:

| Pillar | Question | Red Flag |
|---|---|---|
| Policy | Do we understand ACO REACH benchmark methodology and quality withhold? | Contract requires data we cannot produce |
| Economics | At what medical cost trend does this contract generate positive margin? | Break-even requires cost reduction we cannot achieve |
| Technology | Does our EHR support attribution, care gap identification, and quality reporting? | Attribution data not available in our system |
| Clinical | Which care management programs will reduce our highest-cost utilization? | No care management program exists today |
| Equity | Do our attributed patients have SDOH needs that require non-clinical investment? | SDOH programs not funded in contract |
| Operations | Does our revenue cycle handle the new payment flows? Can we credential care coordination staff? | 90-day credentialing backlog |

### Decision Type 2: Should We Invest in This Digital Health Tool?

A hospital system is evaluating a remote patient monitoring (RPM) platform for cardiovascular patients:

| Pillar | Question |
|---|---|
| Policy | What are the CMS reimbursement codes (CPT 99453–99458)? What is the telehealth policy trajectory? |

| Pillar | Question |
| --- | --- |
| Economics | What is the ROI at our panel size and reimbursement rates? |
| Technology | Does the platform integrate with our EHR via FHIR? |
| Clinical | What is the evidence base for RPM in cardiovascular disease? |
| Equity | Does the platform serve rural patients with broadband and device barriers? |
| Operations | What clinical staff capacity is required to monitor alerts? What are the vendor credentialing requirements? |

### Decision Type 3: Should Our State Design a Global Budget?

A state Medicaid director is considering a global budget proposal:

| Pillar | Question |
| --- | --- |
| Policy | What federal waiver authority is required? What does CMMI require? |
| Economics | How is the benchmark set? What is the growth rate cap? How are hospitals compensated if they underspend? |
| Technology | Does our all-payer claims database have the completeness and timeliness required? |
| Clinical | What care delivery investments are required to bend the cost curve? |
| Equity | Does the budget design include equity adjustors? Who bears the risk for high-need populations? |
| Operations | Do hospitals have the revenue cycle capability to operate under a global budget? |

---

## Key Concepts Introduced in This Chapter

> **Six-Pillar Framework:** HTR's analytical structure for understanding healthcare transformation across Policy, Economics, Technology, Clinical Innovation, Health Equity, and Operations simultaneously.

> **Value-based care (VBC):** Any payment arrangement that ties reimbursement to quality, outcomes, or total cost of care rather than volume of services delivered.

> **Accountable care organization (ACO):** A network of providers that collectively takes accountability for the cost and quality of care for a defined patient population under a VBC contract.

> **FHIR (Fast Healthcare Interoperability Resources):** The HL7 standard for exchanging healthcare information electronically via RESTful APIs; the foundation of modern health data interoperability.

> **HEDIS:** Healthcare Effectiveness Data and Information Set; 90+ quality measures maintained by NCQA used to evaluate health plan and provider performance.

> **HEROI:** Health Equity Return on Investment; an equity-weighted cost-effectiveness metric that gives additional value to interventions targeting disadvantaged populations.

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# CHAPTER 3: Measuring Transformation — The Health Transformation Index

## The Need for a Composite Measure

How do you know if healthcare transformation is working?

This is a surprisingly difficult question. Healthcare improvement can be measured at many levels — the outcomes of individual patients, the quality performance of individual providers, the financial sustainability of individual organizations, the coverage rates of individual states. But none of these tell you whether the system is transforming at the structural level: whether payment reform is actually spreading, whether technology interoperability is actually improving, whether care delivery is actually redesigning.

The Health Transformation Index (HTI) is HTR's answer to this question. It is a composite metric that tracks the overall state of U.S. healthcare transformation across all six pillars, updated quarterly, producing:

- A **national composite score** (0–100)
- **Six pillar sub-indexes** (0–100 each)
- **State Performance Index scores** for all 50 states

The HTI is not a scorecard for individual organizations. It is a macro-level tracking instrument — a barometer for the healthcare system as a whole.

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## HTI Score Interpretation

```
┌──────────────────────────────────────────────────────────────────┐
│                    HTI SCORE INTERPRETATION                      │
├───────────────────┬──────────────────────────────────────────────┤
│  85 – 100         │  Transformative leadership                   │
│                   │  Sector is materially restructured           │
├───────────────────┼──────────────────────────────────────────────┤
│  70 – 84          │  Advanced progress                           │
│                   │  Significant structural change underway      │
├───────────────────┼──────────────────────────────────────────────┤
│  55 – 69          │  Moderate progress                           │
│                   │  Meaningful initiatives, incomplete adoption │
├───────────────────┼──────────────────────────────────────────────┤
│  40 – 54          │  Early stage                                 │
│                   │  Foundational activity, limited penetration  │
├───────────────────┼──────────────────────────────────────────────┤
│  0 – 39           │  Pre-transformation                          │
│                   │  Minimal structural change evident           │
└───────────────────┴──────────────────────────────────────────────┘
```

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## The Architecture of the HTI

The HTI is constructed from the bottom up: individual data elements → component scores → pillar sub-indexes → weighted composite.

```
HTI National Composite (0–100)
│
├── Policy Sub-Index          (20% weight)
│   ├── APM Penetration Rate                       (30%)
│   ├── Medicaid Expansion Status                  (20%)
│   ├── Price Transparency Compliance              (20%)
│   ├── Prior Authorization Reform                 (15%)
│   └── State Innovation Activity                  (15%)
│
├── Economics Sub-Index       (25% weight)
│   ├── VBC Contract Penetration                   (35%)
│   ├── Hospital Operating Margin Stability        (20%)
│   ├── Healthcare Spend Growth vs. GDP            (25%)
│   └── APM Shared Savings Performance             (20%)
│
├── Operations Sub-Index      (10% weight)
│   ├── Administrative Overhead Rate               (35%)
│   ├── Revenue Cycle Friction Index               (35%)
│   └── Workforce Operational Stability            (30%)
│
├── Technology Sub-Index      (15% weight)
```

```
│    ├── FHIR API Adoption Rate                      (30%)
│    ├── EHR Interoperability Score                  (25%)
│    ├── Telehealth Penetration                      (25%)
│    └── AI Clinical Tool Deployment                 (20%)
│
├── Clinical Sub-Index        (20% weight)
│    ├── Quality Composite (HEDIS/Stars)             (35%)
│    ├── Preventable Hospitalization Rate            (25%)
│    ├── Hospital Readmission Rate                   (20%)
│    └── Workforce Sufficiency Index                 (20%)
│
└── Equity Sub-Index          (10% weight)
     ├── Racial Disparity Index                      (35%)
     ├── Rural Access Score                          (25%)
     ├── SDOH Investment Rate                        (20%)
     └── Uninsured Rate by Subgroup                  (20%)
```

---

## Pillar Weights: The Rationale

The pillar weights were established through a structured methodology review and reflect several considerations:

**Economics (25%)** is the highest-weighted pillar because financial incentives are the most powerful lever for system behavior change. Whether payment reform is actually spreading — whether VBC penetration is increasing, whether shared savings are being generated, whether healthcare spend growth is decelerating — is the most sensitive indicator of whether transformation is real.

**Policy (20%) and Clinical (20%)** are equal second because they represent the two fundamental dimensions of transformation: the rules that enable it (Policy) and the outcomes it aims to produce (Clinical). A system could have excellent economics without clinical improvement (cost cutting rather than value creation), or clinical improvement without economic sustainability.

**Technology (15%)** is weighted as an enabler, not an end in itself. FHIR adoption, EHR interoperability, and telehealth penetration matter because they enable the other pillars — not because technology investment is inherently transformative.

**Operations (10%) and Equity (10%)** are the newest and most data-limited pillars. Operations was added to the HTI in the 2026 methodology revision; Equity's weight was reduced from 15% pending improved national SDOH and algorithmic bias data coverage. Both are scheduled for methodology review in 2027.

---

## The Composite Formula

The HTI composite is calculated as:

```
HTI = (Policy × 0.20) + (Economics × 0.25) + (Operations × 0.10)
    + (Technology × 0.15) + (Clinical × 0.20) + (Equity × 0.10)
```

Each component is normalized to a 0–100 scale before aggregation:

```
Normalized Score = ((Raw Value − Minimum Benchmark)
                 / (Maximum Benchmark − Minimum Benchmark)) × 100
```

Where:

- **Minimum Benchmark** = the value representing pre-transformation baseline (score of 0)
- **Maximum Benchmark** = the value representing leading-edge transformation (score of 100)

---

## Data Sources: Where the Numbers Come From

The HTI draws from the following primary federal data sources:

| Source | What It Provides | Update Frequency |
|---|---|---|
| CMS Quality Payment Program | APM participation rates, MIPS scores | Annual |
| CMS Innovation Center (CMMI) | Model participation, shared savings data | Annual |
| CMS Medicaid.gov | Medicaid expansion, enrollment data | Monthly |
| AHRQ HCUP | Preventable hospitalizations, readmissions | Annual |
| ONC Health IT Dashboard | EHR adoption, interoperability measures | Annual |
| HRSA workforce data | Primary care supply, HPSA designations | Annual |
| CDC BRFSS | Behavioral risk factors, preventive care | Annual |
| Census Bureau ACS | Income, insurance, SDOH demographics | Annual |
| KFF State Health Facts | Insurance coverage, policy tracking | Monthly |
| NCQA Quality Compass | HEDIS measures, plan quality data | Annual |

**Data lag and currency adjustment:** Most federal sources have 12–18 month reporting lags. HTR applies a currency adjustment that weights the most recent available year at 100%, data from two years prior at 85%, and three years prior at 70%. This prevents the index from being dominated by stale data.

**Proprietary HTR scoring:** Some components — State Innovation Activity, Payer Innovation Leadership, and Health IT Investment — are scored using HTR's proprietary methodology based on analyst assessment, disclosed data, and surveys. These components undergo formal methodology committee review annually.

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## The National HTI: Quarterly Trajectory

Vermont's national composite score of **82/100** as of Q1 2026 places it at the high end of the "Advanced progress" tier. The quarterly trajectory since 2020:

```
HTI NATIONAL COMPOSITE — QUARTERLY TREND (2020–2026)
─────────────────────────────────────────────────────
100 │
    │
 80 │                                        ●── 82 (Q1 2026)
    │                              ●──●───●──
 60 │               ●───●───●───●──
    │        ●───●──
 40 │    ●──
    │
 20 │
    │
  0 └───────────────────────────────────────────────────
       2020  2021  2022  2023  2024  2025  2026

Key inflection points:
→ 2021: ARP telehealth extensions, MSSP track expansions
→ 2023: CMS Interoperability & PA Final Rule, FHIR mandates
→ 2024: ACO REACH full implementation, price transparency compliance
→ 2025: AHEAD Model expansion, AI deployment acceleration
```

The trend shows consistent improvement, with acceleration after the 2023 FHIR mandate and the 2025 AHEAD Model expansion.

---

## The State Performance Index

Each state receives a Performance Index across six dimensions:

| Dimension | What It Measures |
|---|---|
| Performance Score | Weighted composite of all dimensions |
| Cost Index | Per-capita healthcare spend vs. national median (lower = better) |
| Quality Score | HEDIS + Stars composite for commercial and Medicaid populations |
| Access Score | Uninsured rate, primary care supply, telehealth adoption |
| Equity Score | Racial disparity index + rural access + uninsured by subgroup |
| Innovation Score | APM penetration + VBC contracts + CMMI model participation |

**State performance comparison (Q1 2026):**

| State | Perf. Score | Cost Index | Quality Score | Access Score | Equity Score | Innovation Score |
|---|---|---|---|---|---|---|
| Vermont | 82 | 1.18 | 78 | 73 | 81 | 85 |
| Massachusetts | 79 | 1.24 | 82 | 77 | 74 | 84 |
| Minnesota | 76 | 1.09 | 80 | 71 | 71 | 79 |
| Oregon | 74 | 1.07 | 76 | 68 | 75 | 77 |
| Colorado | 72 | 1.11 | 74 | 70 | 68 | 76 |
| National Median | 51 | 1.00 | 61 | 58 | 52 | 54 |
| Alabama | 36 | 0.94 | 48 | 34 | 28 | 33 |
| Mississippi | 31 | 0.89 | 42 | 28 | 22 | 29 |

```
Note: Cost Index >1.0 means above national median per-capita spend.
```

The striking finding in this data is the inverse correlation between cost and performance in many states. Vermont spends 18% above the national median per capita — yet leads the nation on Innovation Score and performs well above median on Equity. Mississippi spends 11% *below* the national median — yet has near-last performance scores across every dimension. High spending does not produce high performance; but low spending in low-capacity states produces poor performance at low cost.

---

## What the HTI Is Not

The HTI is designed to be useful, not comprehensive. Understanding its limitations is as important as understanding its construction:

**Not a financial health indicator.** Vermont's HTI score of 82 coexists with a hospital sector in acute financial crisis — nine of fourteen hospitals lost money in 2023. The index measures transformation infrastructure maturity, not financial sustainability.

**Not a quality ranking.** The HTI captures adoption of transformation mechanisms, not ultimate health outcomes. A state can score high by adopting APMs that have not yet produced outcome improvements.

**Not an individual organization assessment.** The HTI applies to states and the national system, not to individual health systems, payers, or providers.

**Not current.** Most federal data sources have 12–18 month lags. The HTI's "current" score reflects conditions that existed 6–18 months ago in many components.

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## Trend Analysis: Where Transformation Is Accelerating and Where It Is Stalling

Analyzing the HTI's pillar sub-indexes over time reveals important patterns:

**Accelerating dimensions:**

- **Technology (FHIR adoption):** The 2023 CMS Interoperability and Prior Authorization Final Rule accelerated FHIR API implementation significantly. FHIR adoption has been the fastest-improving component of the HTI since 2023.
- **Operations (revenue cycle modernization):** The shift to electronic prior authorization, driven by the same 2023 rule, is measurably reducing revenue cycle friction.

**Stable, improving dimensions:**

- **Clinical (quality composite):** HEDIS and Stars scores improve incrementally year-over-year, reflecting sustained investment in quality improvement programs.
- **Economics (VBC penetration):** VBC contract penetration has grown from approximately 40% of commercially insured in 2020 to approximately 58% in 2025, but the pace has moderated.

**Stalled dimensions:**

- **Equity (racial disparity index):** Racial health disparities have not materially narrowed in the U.S. despite increased policy attention and equity-focused program investments. This is the most troubling trend in the HTI and reflects the inadequacy of clinical interventions in the absence of SDOH investment.
- **Policy (prior authorization reform):** Despite federal rules and significant state legislation, prior authorization burden remains high. Implementation has lagged far behind legislation.

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## Key Concepts Introduced in This Chapter

> **Health Transformation Index (HTI):** HTR's composite metric tracking U.S. healthcare transformation progress across six pillars, produced quarterly at national and state levels.

> **APM Penetration Rate:** The percentage of total healthcare spending flowing through alternative payment models with two-sided risk; the most heavily weighted component of the Economics sub-index.

> **State Innovation Activity:** HTR's proprietary score assessing each state's portfolio of health transformation programs, including waivers, global budget participation, and innovation grants.

> **Currency adjustment:** HTR's methodology for accounting for data lag in federal sources by weighting more recent data more heavily.

> **Prevention Quality Indicators (PQI):** AHRQ's set of measures identifying preventable hospitalizations — admissions for ambulatory-care-sensitive conditions that adequate primary care should prevent.

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*End of Part I*

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